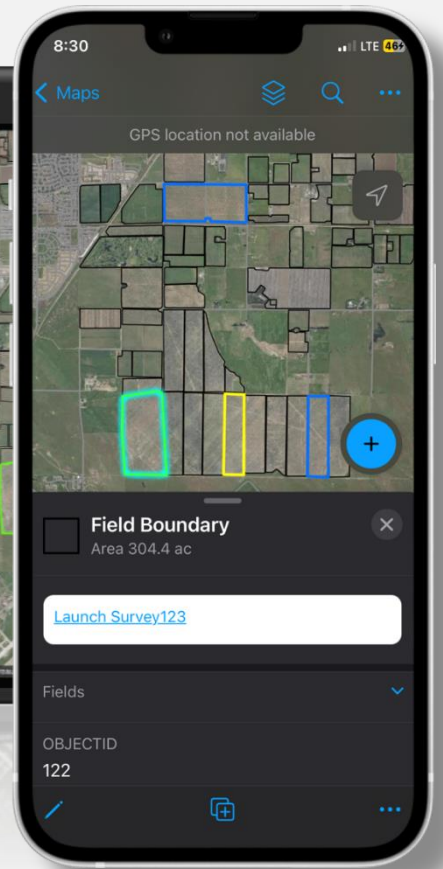
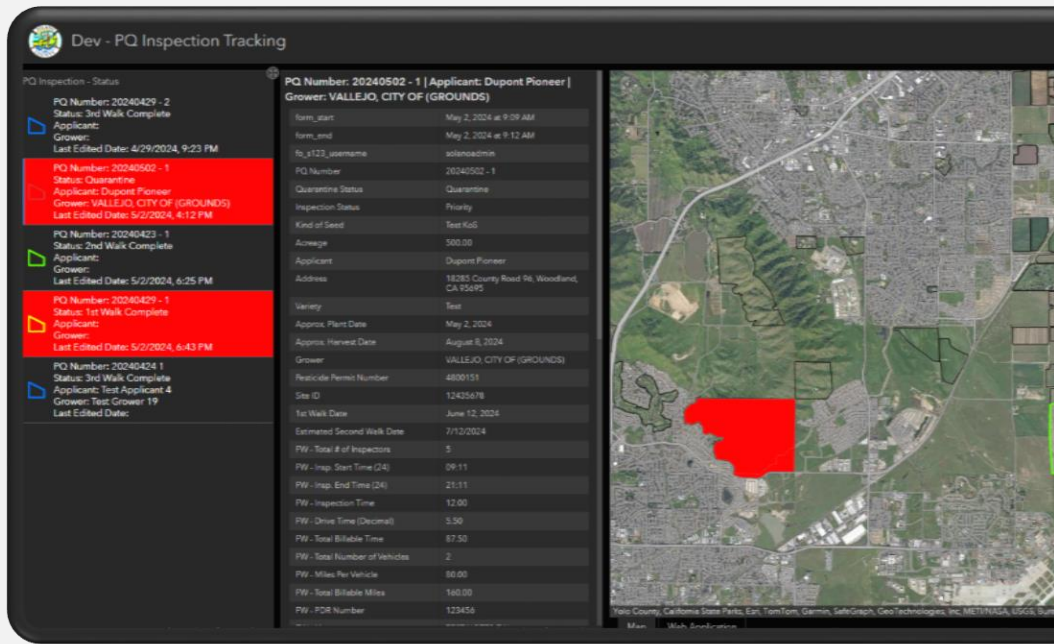




TECHNICAL DOCUMENTATION

PQ Inspection Tracking

Prepared for: Solano County, California
Solano County GIS

Prepared by: KCI Technologies, Inc.





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DOCUMENT CONTROL

Version	Date	Purpose / Action	Author
1.0	5/3/2024	Document created	Andrew Blowers (KCI)
1.1	2/27/2025	Updating PQ Inspection Tracking App	Darbie Gibbs (KCI)
1.2	5/9/2025	Adding Survey123 Webform Compatibility	Darbie Gibbs (KCI)



OVERVIEW

Stakeholders: Solano County, California – Solano County GIS

Points of Contact:

IT/GIS POC: Daniel Machado

TECHNICAL OVERVIEW

The solution can be summarized at a high level as follows. Subsequent sections will provide greater detail for the parts described here.

1. Web Map
2. Web / Desktop Application
3. Dashboard
4. Survey123

DATA

Data for this solution is hosted and stored within the Solano Enterprise Portal.

ArcPro Project – Project supporting the solution

<\\gis.solanocountygis.local\E\ServiceUpdates\PQ Inspection Tracking\PQ Inspection Tracking\PQ Inspection Tracking.aprx>

Mapserver: PQ_Inspection_Tracking_Views – Includes views Quarantine/Inspection Status & All Records

https://solanocountygis.com/server/rest/services/PQ_Inspection_Tracking/PQ_Inspection_Tracking_Views/MapServer

pq_inspection_tracking – Table that supports the survey123

https://solanocountygis.com/server/rest/services/PQ_Inspection_Tracking/PQ_Inspection_Tracking/FeatureServer/1

Field Boundaries – All PQ Inspection Tracking records are created via field boundary polygons.

https://solanocountygis.com/server/rest/services/PQ_Inspection_Tracking/PQ_Inspection_Tracking/FeatureServer/0

Road Centerlines – Reference layer of County road centerlines maintained by various County resources.

https://services2.arcgis.com/SCn6czzcqKAFwdGU/ArcGIS/rest/services/Road_Centerlines/FeatureServer/0

Solano County Boundary – Standard Solano County boundary feature layer

https://services2.arcgis.com/SCn6czzcqKAFwdGU/arcgis/rest/services/County_Boundary/FeatureServer

grower_pesticide_perm_num.csv – Supports the survey123 form

<https://solanocountygis.com/portal/home/item.html?id=86638bb163714fc18641563b490b0b37>

applicant_address.csv - Supports the survey123 form

<https://solanocountygis.com/portal/home/item.html?id=eec7da339074223b24d9194deb7007e>



commodity_walk_days.csv – Supports the survey123 form

<https://solanocountygis.com/portal/home/item.html?id=4cade65c246d48a8848af0810c9dd653>

SURVEY 123 AND FIELDMAP

Users can launch a Survey123 form from Field Maps, Dashboard, or Web / Desktop Application. Forms started from Field Maps must be launched via the URL on the Field Boundaries feature layer. Subsequent edits to forms must be completed via the Inbox directly in the Survey123 desktop or mobile field app.

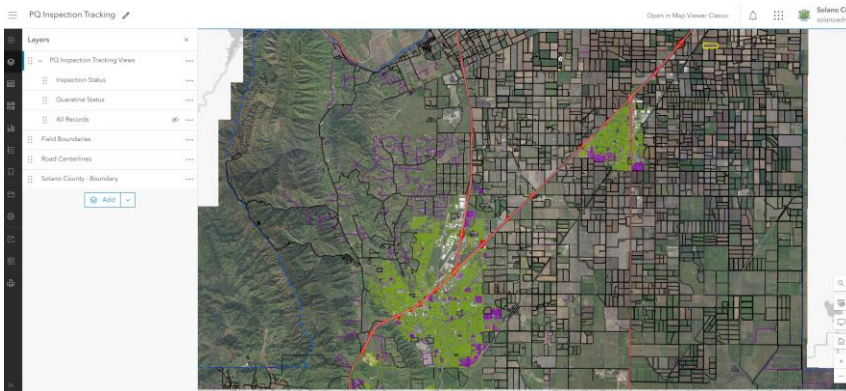
\\gis.solanocountygis.local\E\ServiceUpdates\PQ Inspection Tracking\PQ Inspection Tracking PRD Survey123 Form.zip

CSV Files: **grower_pesticide_perm_num.csv**

applicant_address.csv

commodity_walk_days.csv

WEBMAP



<https://solanocountygis.com/portal/home/item.html?id=73ef5034cdb84bd696b4adc883fafd3a>

Group / Layer Name	Description
Field Boundaries	Reference feature layer of all relevant fields for PQ Inspections. While a reference feature, users can add new fields via the Edit widget in the Web / Desktop Application or Field Maps Mobile Application.
Road Centerlines	Reference layer of County road centerlines maintained by various County resources.
County Boundary	Standard Solano County boundary feature layer
PQ Inspection Tracking Views (Layer Group)	Note that all layers in this group are dynamic and are updated when changes occur to individual PQ records.
Inspection/Quarantine Status	A view aggregating the PQ Inspection Tracking records related to field boundary features and grouping them by the inspection and quarantine status. The layer's purpose is to consolidate both statuses into a single field. The layer is configured to filter out records with a "Complete" status and those with a "walk" date or created date outside of the current calendar year. Standby



	• Priority • 1st Walk Complete • 2nd Walk Complete • 3rd Walk Complete • Hold • Quarantine
All Records	Layer that shows all PQ records. PQ records are displayed with the field boundary geometry they are related to. Expect stacked geometry for these layers if more than one PQ record exists for a field – <i>each</i> PQ record is symbolized with its related geometry.
PQ Inspection Tracking (S123 Submissions) (Table)	This is the database table to which the Survey123 submissions are written. The table facilitates the views and display of records in the dashboard. It is not filtered by any date parameters and will include all submissions from the PQ Inspection Tracking form.

Pop-Up Configuration:

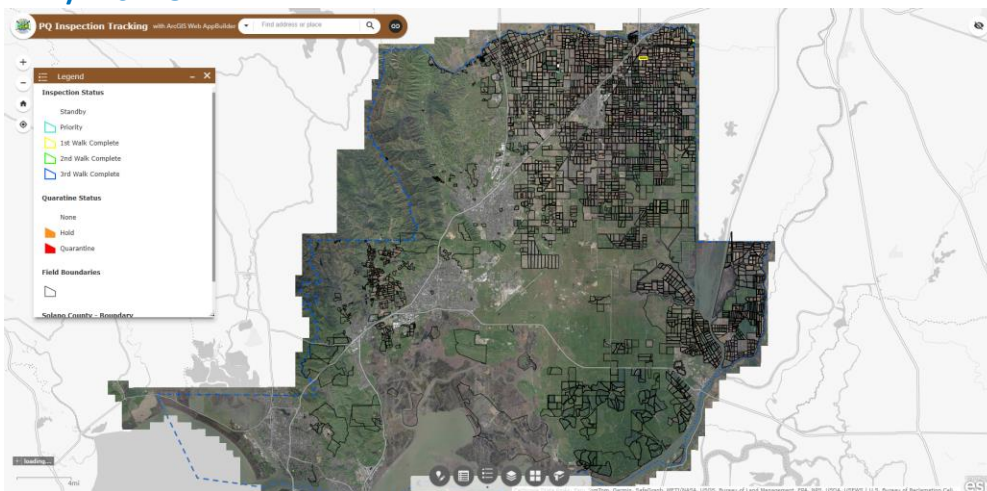
Pop ups are configured so when you click on a parcel you can access the Inspection survey for that parcel. Pop ups are configured using html to embed survey link. Pop up buttons passes parameters into the survey.

1. **incoming_id** → {globalid}

HTML:

```
<p>  
  
<a href="arcgis-survey123://?itemID=2cec2a099a764b928fe343334569d4f4&field:incoming_id={GLOBALID}">Launch Survey123</a>  
  
</p>
```

WEB/DESKTOP APP



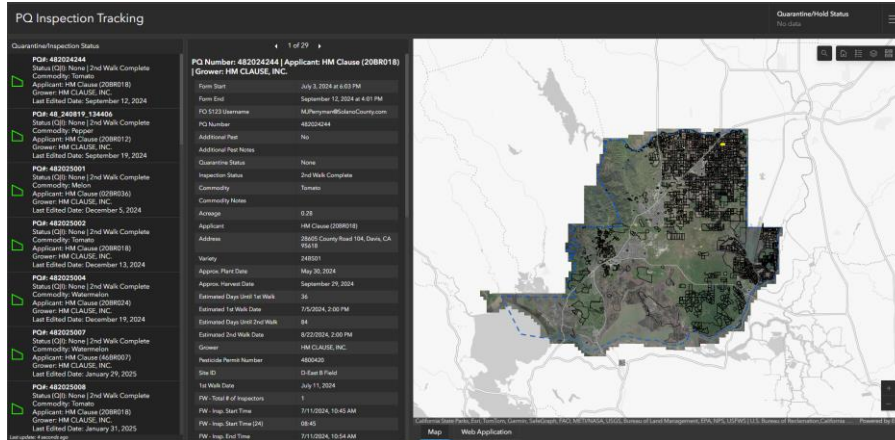
<https://solanocountygis.com/portal/apps/webappviewer/index.html?id=00ba255a6c3e48ee9e59c060aa55f8d7>



Features:

1. Search bar
 - Search by address, PQ Number, Applicant or Grower
2. Widget Launch Bar
 - Attribute Table – allows users to view and export the attribute table for all relevant features in the layer list.
 - Legend – displays the visible layers in the map.
 - Layer List – allows the user to toggle layers on/off, change order, adjust transparency, and other functions.
 - Basemap Gallery – allows users to change the basemap utilizing the organization's complete base map gallery
 - Bookmarks – allows users to store map bookmarks in the application.
 - Note that these bookmarks will not carry over to the Field Maps application.

DASHBOARD



<https://solanocountygis.com/portal/apps/dashboards/27edc16d7c314a5581b93e9bc6a32d3b>

Features:

1. Left side of the dashboard – PQ Inspection / Field status list and individual PQ records
 - The “Status” list allows for the selection of multiple records and controls the individual record list to the right.
2. Right side of the dashboard – map / tools, web / desktop application (tab)
 - The “Status” list is controlled by the map's extent.
 - Selecting a field boundary polygon from the map displays the feature’s pop-up, which contains the Survey123 hyperlink used to launch a new form in the desktop application.

ENHANCEMENTS

Updating PQ Inspection Tracking - 2/27/2025

Request: Increase Character Limit for "Additional Pest Notes" Field. Increase the character limit for this field from 255 characters to 2,000 characters or more.

In pro: Change field length in pro project. Republish.

In Survey123: Update esrilenlength to same character limit.



Request: Using a csv file, have "Contact Name & Phone Number" auto-populate based on the selection of the Applicant. Currently only the address auto-populates

Updated the Survey123: Survey123 formula: pulldata('applicant_address1', 'contactname', 'name', \${applicant})

In webmap: Configured the table popups to contain and reorder the new attributes.

In webapp: Configured the attribute table to reorder the new attributes.

*Got new updated contact CSV from Cody Zacharia - 4/3/2025

Request: Add a Lead Inspector Field

- For the first, second, and third walks, add a new data entry field above the "# of Inspectors" section.
- This field should allow up to 150 characters and will be used to enter the full name of the Lead Inspector.

In pro: Added lead inspector fields for each walk to the pq_inspection_tracking table.

☐	fw_lead_inspector	FW - Lead Inspector	Text	☒	☐						150
☐	sw_lead_inspector	SW - Lead Inspector	Text	☒	☐						150
☐	tw_lead_inspector	TW Lead Inspector	Text	☒	☐						150

In Survey123: Addes Fields, Added esri field alias field. (FW - Lead Inspector, SW - Lead Inspector, TW - Lead Inspector)

In webmap: Configure the table popups to contain and reorder the new attributes.

In webapp: Configure the attribute table to reorder the new attributes.

Request: Add a data field called "Additional Time" to the first, second, and third walk pages of the survey to capture time spent on smaller tasks related to the inspection of a PQ field outside of the actual inspection of the field.

In Pro: Updated fields in pq_inspection_tracking table to include & republish:

☐	sw_additional_time	SW - Additional Time (Decimal)	Double	☒	☐	Numeric					38	8
--------------------------	--------------------	--------------------------------	--------	-------------------------------------	--------------------------	---------	--	--	--	--	----	---

In survey 123 Connect: Updated xls form to include line 50 fw additional time field, line72 sw additional time field, and line 94 tw additional field. Updated the Total billable hours calculation in each 3 walks to include additional time.

$\${fw_additional_time} + ((\${fw_insp_time} + \${fw_drive_time}) * \${fw_total_number_inspectors})$

Added esri field alias field. (FW - Additional Time (Decimal), SW - Additional Time (Decimal), TW - Additional Time (Decimal))

In webmap: Configure the table popups to contain and reorder the new attributes.

In webapp: Configure the attribute table to reorder the new attributes.

Adding Survey123 Webform Compatibility - 5/9/2025

Request: Can we please update the pop-up for the PQ Inspection Tracking so that it links to the web version of Survey123 not just the mobile version?

In Web Map:

Reworked the HTML in the web map pop-up to include second link with the option to open in browser.

*Version=3.21 was added to URL to work around current V 3.22 bug breaking Survey123 webform. App version not affected.



```
<p>
  <a target="_blank" rel="noopener noreferrer" href="https://survey123.arcgis.com/share/2cec2a099a764b928fe343334569d4f4?portalUrl=https://solanocountygis.com/portal&version=3.21&field:incoming_id={GLOBALID}">Open Survey123 in Web Browser&nbsp;</a>
<br>
  <br>
  <a href="arcgis-survey123://?itemID=2cec2a099a764b928fe343334569d4f4&portalUrl=https://solanocountygis.com/portal&field:incoming_id={GLOBALID}">Open in Survey123 Mobile/Desktop App</a>
</p>
```

CSV:

The header fields in the Commodity_Walk_Days.csv were updated so not to use capitals or spaces in the header name. They are not compatible with the webform version.

In Survey 123:

Form was updated as follows to make compatible with webform version of survey123:

- Fieldboundary_guid:
 - o Calculation was modified to include brackets. \${incoming_id}
- Form_start, form_end, fo_s123_username, form_version:
 - o Moved from outside to inside group with "field-list" appearance. Questions must be inside a group with field-list appearance when the form type is set to "Pages" otherwise it will break in webform.
- Fw_est_days, sw_est_days:
 - o Updated the calculation formula to include the new csv lowercase field headers.
 - o pulldata('commodity_walk_days_dev', 'firstwalkestimate', 'name', \${commodity})
 - o pulldata('commodity_walk_days_dev', 'secondwalkestimate', 'name', \${commodity})
- fw_insp_time_calc, sw_insp_time_calc, tw_insp_time_calc:
 - o Calculation formula updated to make compatible with webform.
 - o round((decimal-date-time(\${fw_insp_end_time}) - decimal-date-time(\${fw_insp_start_time})) * 24, 2)
- fw_total_billable_time, sw_total_billable_time, tw_total_billable_time:
 - o Calculation formula updated to make compatible with webform.
 - o coalesce(\${fw_additional_time}, 0) + ((coalesce(\${fw_insp_time}, 0) + coalesce(\${fw_drive_time}, 0)) * coalesce(\${fw_total_number_inspectors}, 0))
- fw_total_billable_miles, sw_total_billable_miles, tw_total_billable_miles:
 - o Calculation formula updated to make compatible with webform.
 - o coalesce(\${fw_miles_per_vehicle}, 0) * coalesce(\${fw_total_number_of_vehicles}, 0)



APPENDIX

NOTE

The following appendices memorialize solution scripts and configurations to be used solely as reference. If stored outside of ArcGIS Enterprise (for JSON) / F drive (for Python script), actual code should be stored in and leverage from a code repo. If a restore is required, use the code and/or files provided with this document.



Appendix A – View Layers

Appendix A-1: SQL Code – v_pq_inspection_tracking_all

The following is a copy of the SQL code for the v_pq_inspection_tracking_all.

```
SELECT pq.objectid,  
       pq.globalid,  
       pq.form_start,  
       pq.form_end,  
       pq.fo_s123_username,  
       pq.form_version,  
       pq.fieldboundary_guid,  
       pq.pq_number,  
       pq.quarantine_status,  
       pq.inspection_status,  
       pq.acreage,  
       pq.applicant,  
       pq.address,  
       pq.contactname,  
       pq.phonenumber,  
       pq.variety,  
       pq.approx_plant_date,  
       pq.approx_harvest_date,  
       pq.grower,  
       pq.pesticide_permit_number,  
       pq.site_id,  
       pq.first_walk_date,  
       pq.fw_lead_inspector,  
       pq.fw_total_number_inspectors,  
       pq.fw_insp_start_time_m,  
       pq.fw_insp_end_time_m,
```



pq.fw_insp_time,
pq.fw_drive_time,
pq.fw_additional_time,
pq.fw_total_billable_time,
pq.fw_total_number_of_vehicles,
pq.fw_miles_per_vehicle,
pq.fw_total_billable_miles,
pq.fw_pdr_number,
pq.first_walk_notes,
pq.second_walk_date,
pq.sw_lead_inspector,
pq.sw_total_number_inspectors,
pq.sw_insp_start_time_m,
pq.sw_insp_end_time_m,
pq.sw_insp_time,
pq.sw_drive_time,
pq.sw_additional_time,
pq.sw_total_billable_time,
pq.sw_total_number_of_vehicles,
pq.sw_miles_per_vehicle,
pq.sw_total_billable_miles,
pq.sw_pdr_number,
pq.second_walk_notes,
pq.third_walk_date,
pq.tw_lead_inspector,
pq.tw_total_number_inspectors,
pq.tw_insp_start_time_m,
pq.tw_insp_end_time_m,



pq.tw_insp_time,
pq.tw_drive_time,
pq.tw_additional_time,
pq.tw_total_billable_time,
pq.tw_total_number_of_vehicles,
pq.tw_miles_per_vehicle,
pq.tw_total_billable_miles,
pq.tw_pdr_number,
pq.third_walk_notes,
pq.created_date,
pq.created_user,
pq.last_edited_date,
pq.last_edited_user,
pq.incoming_id,
pq.commodity,
pq.commodity_other_notes,
pq.additional_pest,
pq.additional_pest_notes,
pq.fw_est_date,
pq.sw_est_date,
pq.fw_est_days,
pq.sw_est_days,
pq.fw_insp_start_time,
pq.fw_insp_end_time,
pq.sw_insp_start_time,
pq.sw_insp_end_time,
pq.tw_insp_start_time,
pq.tw_insp_end_time,



```
fb.objectid AS fb_objectid,  
fb.globalid AS fb_globalid,  
fb.shape  
FROM pq_inspection_tracking_evw pq  
JOIN fieldboundaries_evw fb ON pq.fieldboundary_guid::text = fb.globalid::text;
```

Appendix A-2: SQL Code – v_pq_inspection_tracking_status

The following is a copy of the SQL code for the v_pq_inspection_tracking_status.

```
SELECT pq.objectid,  
pq.pq_number,  
pq.commodity,  
pq.applicant,  
pq.grower,  
pq.inspection_status,  
pq.quarantine_status,  
pq.first_walk_date,  
pq.second_walk_date,  
pq.third_walk_date,  
pq.created_user,  
pq.created_date,  
pq.last_edited_user,  
pq.last_edited_date,  
pq.incoming_id,  
pq.fieldboundary_guid,  
fb.shape,  
CASE  
    WHEN pq.inspection_status::text = 'Standby'::text AND pq.quarantine_status::text = 'None'::text THEN  
        'Standby'::text  
    WHEN pq.inspection_status::text = 'Priority'::text AND pq.quarantine_status::text = 'None'::text THEN  
        'Priority'::text
```




```
WHEN pq.inspection_status::text = '1st Walk Complete'::text AND pq.quarantine_status::text = 'None'::text THEN
'1st Walk Complete'::text

WHEN pq.inspection_status::text = '2nd Walk Complete'::text AND pq.quarantine_status::text = 'None'::text THEN
'2nd Walk Complete'::text

WHEN pq.inspection_status::text = '3rd Walk Complete'::text AND pq.quarantine_status::text = 'None'::text THEN
'3rd Walk Complete'::text

WHEN pq.inspection_status::text = 'Standby'::text AND pq.quarantine_status::text = 'Hold'::text THEN 'Hold'::text

WHEN pq.inspection_status::text = 'Priority'::text AND pq.quarantine_status::text = 'Hold'::text THEN 'Hold'::text

WHEN pq.inspection_status::text = '1st Walk Complete'::text AND pq.quarantine_status::text = 'Hold'::text THEN
'Hold'::text

WHEN pq.inspection_status::text = '2nd Walk Complete'::text AND pq.quarantine_status::text = 'Hold'::text THEN
'Hold'::text

WHEN pq.inspection_status::text = '3rd Walk Complete'::text AND pq.quarantine_status::text = 'Hold'::text THEN
'Hold'::text

WHEN pq.inspection_status::text = 'Standby'::text AND pq.quarantine_status::text = 'Quarantine'::text THEN
'Quarantine'::text

WHEN pq.inspection_status::text = 'Priority'::text AND pq.quarantine_status::text = 'Quarantine'::text THEN
'Quarantine'::text

WHEN pq.inspection_status::text = '1st Walk Complete'::text AND pq.quarantine_status::text = 'Quarantine'::text
THEN 'Quarantine'::text

WHEN pq.inspection_status::text = '2nd Walk Complete'::text AND pq.quarantine_status::text = 'Quarantine'::text
THEN 'Quarantine'::text

WHEN pq.inspection_status::text = '3rd Walk Complete'::text AND pq.quarantine_status::text = 'Quarantine'::text
THEN 'Quarantine'::text

WHEN pq.inspection_status::text = 'Complete'::text THEN 'Complete'::text

ELSE NULL::text

END AS status

FROM pq_inspection_tracking_evw pq

JOIN fieldboundaries_evw fb ON pq.fieldboundary_guid::text = fb.globalid::text;
```